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1 Introduction

Housing prices in Copenhagen have undergone a dramatic transformation over the past three decades. Between the early 1990s recovery, after falling interest rates and renewed monetary stability ended a prolonged housing slump, the pre-2008 boom driven by cheaper credit and new mortgage types, and the post-crisis correction caused by rising risk premiums and stricter lending, the prolonged low-rate environment of the 2010s that fuelled a renewed surge in urban apartment prices, and the sharp monetary tightening of 2022-2023 that abruptly raised borrowing costs and cooled demand, the city's apartment market has consistently been shaped less by neighbourhood-level fundamental characteristics than by the price of credit. Understanding how interest rates feed into housing prices is therefore not only of academic interest, but directly relevant for homeowners using Danish “realkreditlån”, for lenders pricing mortgage bonds, and for policymakers concerned with financial stability.

This paper studies the relationship between Copenhagen housing prices and Danish long-term mortgage bond yields over a roughly 30-year horizon. Because consistent historical data on the actual mortgage interest rate are not available, we use the yield on long-term Danish mortgage bonds as a proxy for the underlying cost of long-term financing.

2 Data

The time series data utilized for this project were sourced from Finans Danmark. Our project consists of two datasets:

- Realized transaction prices for owner-occupied apartments in Copenhagen, measured in DKK per square metre (1992 - 2025). Reported quarterly.
- Long-term mortgage bond yields (*obligationsrenten*), 1997–2026. Reported weekly.

Our primary dataset is a data aggregation of the averaged transaction price for owner-occupied housing in Copenhagen, measured in Danish kroner per square meter (DKK/m²). The series consists of 136 observations covering the period 1992 Q1 to 2025 Q4. Over this period the average cost price rose from

approximately 5,250 DKK/m^2 in 1994 to Q4 to 68,450 DKK/m^2 in 2025 Q4, an increase of roughly 1,200% over three decades.

Figure 2.1 displays the raw data of both the square meter prices in Copenhagen, and the long-term mortgage bond yield over the time period. Three episodes are visible, a strong pre-2008 boom that peaked around 2006-2007, the correction associated with the financial crisis in 2008-2009, and sustained a recovery from 2012 onward, briefly interrupted again around 2020 before resuming growth through 2024. The series shows a clear upward trend with no pronounced seasonal pattern, suggesting the level is non-stationary and requires differencing for time-series modeling. The long-term mortgage bond yield displays a broadly opposite trajectory, declining from approximately 7.5% in the late 1990s to roughly 4% at the end of the sample, with a temporary spike around 2008, a prolonged low-rate period through 2021, and a sharp tightening cycle in 2022-2023. As with housing prices, the level is clearly non-stationary and will likewise require differencing for time-series modelling.

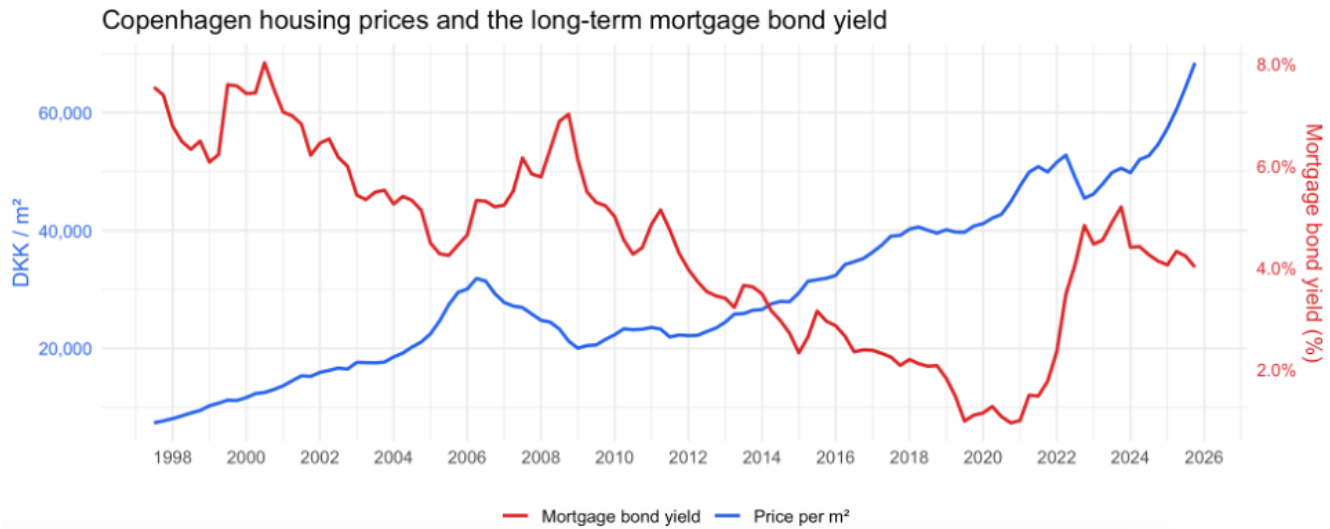
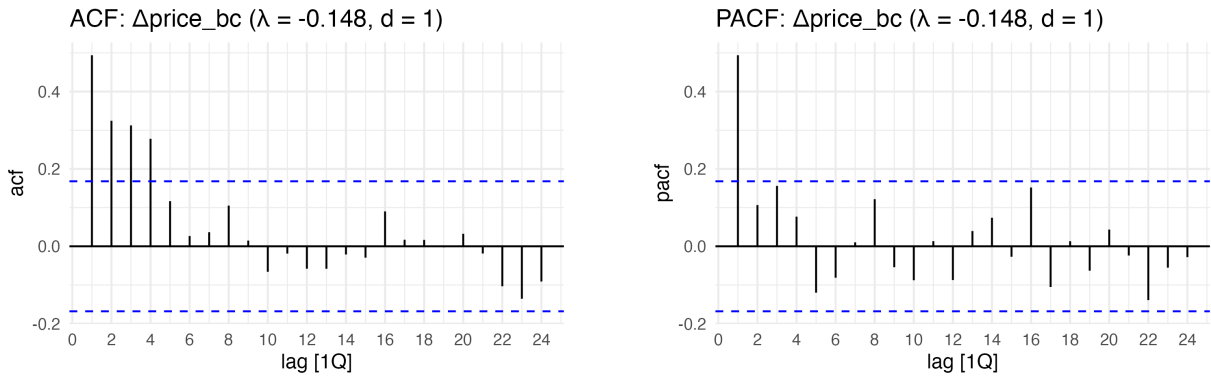


Figure 2.1: Plot of square meter prices in Copenhagen and the long-term mortgage bond yield.

To examine the autocorrelation structure of the first difference, Figure 2.2a and Figure 2.2b presents the ACF and PACF of $\Delta price$. The ACF exhibits significant positive autocorrelation at the first four lags, while the PACF declines more rapidly after the second lag. This pattern is consistent with a low-order ARIMA structure in the differenced series and motivates the ARIMA framework adopted in Section 4.1.



(a) Autocorrelation function (ACF) of square meter prices in Copenhagen. (b) Partial autocorrelation function (PACF) of square meter prices in Copenhagen.

To formally test for stationarity, we apply both the Augmented Dickey-Fuller (ADF) test, where the null hypothesis is the presence of a unit root, and the KPSS test, where the null hypothesis is stationarity. Using two tests with opposite nulls makes the inference clearer when results are borderline. We test the transformed price series at two differencing orders, levels ($d = 0$), and first differences ($d = 1$), and select the smallest d at which both tests agree that the series is stationary.

In levels, both tests indicate non-stationarity for the Box-Cox transformed price, the ADF is -1.54 , well above the 5% critical value of -2.89 , which does not reject the unit root, and the KPSS is 1.95 , well above the 1% critical value of 0.739 . For the long-term mortgage bond yield, the ADF is -0.92 , and the KPSS is 2.11 , both indicating non-stationarity at level. After taking first differences, both tests agree that the series has become stationary. For the differenced transformed price, ADF rejects the unit root at the 1% level (-3.91) and KPSS does not reject stationarity at the 5% level (0.21 , below the 5% critical value of 0.463). For the differenced mortgage bond yield the evidence is unambiguous, ADF is -7.51 and KPSS is 0.05 . We therefore conclude that both the transformed price series and the long-term mortgage bond yield are $I(1)$ and work with first differences in the subsequent modelling.

2.1 Data choices and limitations

The Danish mortgage interest rate is proxied by the long-term Danish mortgage bond yield, the *obligationsrente*, rather than by a directly observed borrower-facing mortgage interest rate, the *realkreditrente*.

This choice reflects the structure of the Danish mortgage market, loans are funded through the issuance of callable mortgage bonds, so the long-term mortgage bond yield captures the underlying market financing cost behind mortgage lending. Although the interest rate for the borrower's mortgage would in principle be the most precise measure of household borrowing costs, it varies between mortgage institutions, loan types, repayment structures and loan-to-value ratios, and consistent historical data covering all relevant components is not available for our 30-year sample. In particular, the final cost faced by borrowers also includes the mortgage contribution fee, the *bidragssats*, an institution- and loan-specific contribution margin paid on top of the bond yield, for which reliable data is only available from 2014 onward. Since 2014, this margin has remained relatively stable at approximately 0.7 – 0.8 percentage point. We therefore use the long-term mortgage bond yield as a consistent market-level proxy for mortgage finance conditions and treat the role of the mortgage contribution fee as a methodological limitation.

Finans Danmark reports three series of data pertaining to the mortgage bond yields: The short euro rate, the short danish rate, and the long danish rate. We use the long Danish rate because long-dated Danish callable mortgage bonds are priced directly off this yield, so it most closely tracks the rate that borrowers face on standard 30-year fixed-rate mortgages, the dominant financing instrument for Danish owner-occupied housing throughout our sample. The short Danish rate is not used because it is primarily relevant for floating-rate loans (F1/F3/F5), which have not been the dominant mortgage product throughout the entire 30-year period and would therefore lack a consistent interpretation across the sample. The short euro rate is excluded because it affects Danish housing prices only indirectly, through its influence on the Danish rates already included.

To combat the difference in reporting (quarterly vs. weekly) in our two datasets, we aggregated the weekly observations to a quarterly frequency by taking the simple within-quarter mean. The mean is preferable to an end-of-quarter value because housing transactions are spread throughout the quarter, so the average cost of credit over the quarter is the economically relevant object for pricing decisions. End-of-period sampling would put disproportionate weight on a single week and introduce noise unrelated to the housing-market signal.

3 Methodology

The empirical strategy is built around three complementary time-series models, each of which captures a different aspect of the relationship between Copenhagen housing prices and Danish long-term mortgage bond yields. The three models are arranged from simplest to most general: a univariate ARIMA serves as a pure forecasting benchmark, an ARIMAX adds the long-term mortgage bond yield as an exogenous driver, and a bivariate VAR allows the two series to interact freely. This progression makes it possible to attribute incremental forecasting power to each layer of information added to the model.

The starting point is the univariate ARIMA model fitted to the Box-Cox transformed price series. Because an ARIMA relies only on past values and past forecast errors of the variable being modelled, it provides a clean benchmark against which any gains from incorporating economic information can be measured. The Box-Cox transformation, motivated by the preliminary analysis, ensures that the variance is approximately stable across the sample, and first differencing is handled internally through the $d = 1$ specification. The autoregressive and moving-average orders are selected by minimizing AICc over a grid search, with residual diagnostics used to confirm that no systematic structure remains in the errors.

The ARIMAX specification extends this baseline by including the long-term mortgage bond yield, used throughout as a proxy for the mortgage interest rate, together with its lags as exogenous regressors. The motivation is theoretical as well as empirical: long-term Danish mortgage bonds are priced directly off this yield, so it captures the dominant channel through which monetary conditions translate into household borrowing costs and, in turn, into housing demand. By embedding this channel inside an ARIMA error structure, the ARIMAX retains the parsimony of the baseline while letting the model react to changes in the cost of credit. A key assumption underlying the ARIMAX is that causality runs in only one direction, from interest rates to prices.

That assumption is relaxed in the VAR model, where housing prices and the long-term mortgage bond yields are treated symmetrically as jointly determined variables. The VAR is estimated on the first differences of the two series, with the lag order selected by AIC subject to a moderate upper bound to control

parameter inflation. The VAR plays a dual role in the analysis, it serves as a forecasting model in its own right, and as a diagnostic for the one-way assumption that underlies the ARIMAX.

The three models are evaluated against one another in an out-of-sample setting. The sample is divided into a training period (1997 Q3 - 2020 Q2, 92 observations) used to estimate all parameters, and a test period (2020 Q3 - 2025 Q4, 22 observations) used to assess forecast accuracy. Performance is summarised using RMSE, MAE, and MAPE, supplemented by AIC and BIC for in-sample fit and by residual diagnostics for model adequacy.

4 Models and results

Building on the stationary and structural-break analysis of Section 2, this section estimates three candidate models on the training sample. Section 4.1 begins with a univariate ARIMA on the Box-Cox transformed price series as a pure time-series benchmark, before extending to an ARIMAX specification that adds the lagged long-term mortgage bond yield as an exogenous regressor in order to isolate the marginal contribution of the rate channel. Section 4.2 then estimates a bivariate VAR on the Box-Cox transformed price series and the rate, which treats both variables as endogenous and permits feedback in either direction.

4.1 ARIMA and ARIMAX

An $ARIMA(p, d, q)$ model expresses the Box-Cox transformed price y_t (with $\lambda = 0.39$) as an autoregressive moving-average process in its first difference ($d = 1$, established in Section 2). The ARIMAX extension augments this by adding the lagged long-term mortgage bond yield as an exogenous regressor in the mean equation, so that price dynamics depend both on their own past and on the cost of credit. The optimal rate lag is selected from $\{0, 1, 2, 4\}$ quarters by minimizing AICc.

4.1.1 Baseline ARIMA

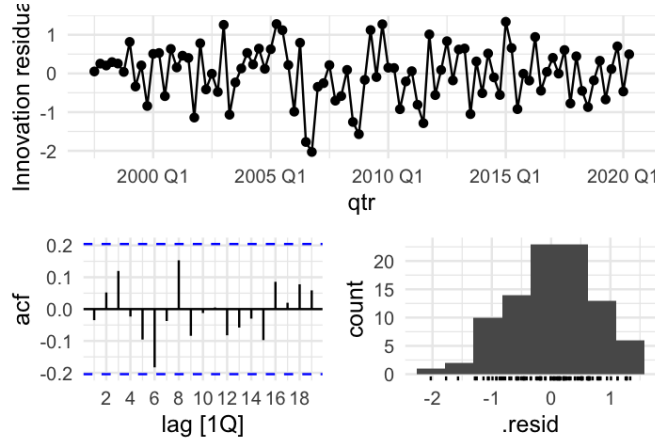


Figure 4.1: Residual plots for baseline ARIMA model.

For the univariate benchmark, an AICc grid search selects an $ARIMA(1, 1, 0)(1, 0, 0)[4]$ specification with a drift on the Box-Cox transformed price. The model differences the series once, captures short-run momentum through a single non-seasonal autoregressive lag, and adds a seasonal autoregressive term at lag 4 (one year) to account for annual persistence in price growth. Letting $y_t = (P_t^\lambda - 1)/\lambda$ denote the Box-Cox transformed price, the fitted equation is

$$\Delta y_t = 0.1590 + 0.5613 \Delta y_{t-1} + 0.2319 \Delta y_{t-4} - 0.1302 \Delta y_{t-5} + \varepsilon_t$$

Where the -0.1302 coefficient at lag 5 arises from the multiplicative interaction between the non-seasonal and seasonal autoregressive terms ($\phi_1 \cdot \Phi_1 = 0.5613 \cdot 0.2319$). The estimated drift of 0.1590 represents the average quarterly increment in the Box-Cox transformed price, on the original DKK/m^2 scale this corresponds to persistent positive growth, consistent with the long-run upward trend visible in Section 2.

4.1.2 ARIMAX with the long-term Mortgage Bond Yield

The ARIMAX specification extends the univariate ARIMA by adding the lagged long-term mortgage bond yield as an exogenous regressor. This allows house-price dynamics to depend not only on their own past but also on the cost of mortgage credit, the economically central driver of Danish housing demand.

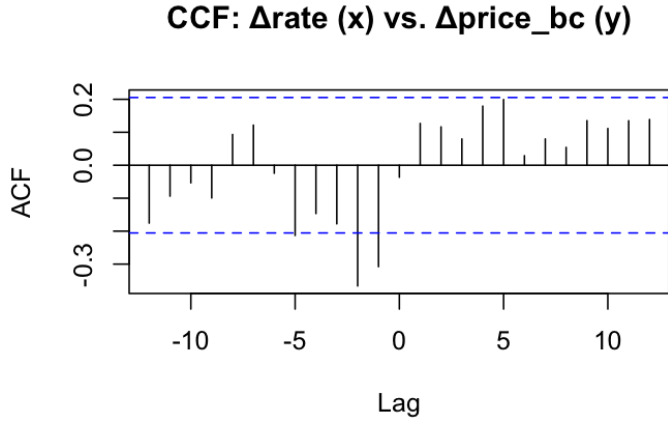


Figure 4.2: Cross-correlation function between the differenced mortgage bond yield ($\Delta rate_t$) and the differenced Box-Cox transformed price (Δy_t).

Lag k (quarters)	AICc
0	219
1	212
2	208
3	215
4	214
5	212

Table 1: AICc values for ARIMAX specifications with the mortgage bond yield at candidate lags k .

Figure 4.2 displays the cross-correlation function between the Δy_t and $\Delta rate_t$. The strongest negative correlation occur at negative lags of one and two quarters, indicating that rate movements lead price movements with a short delay. To formally select the lag at which rate enters the model, we estimate ARIMAX specifications with the rate at lags $k \in \{0, 1, 2, 3, 4, 5\}$ and compare the resulting AICc values (Table 1). The AICc-minimizing specification places the rate at lag $k = 2$ quarters, which we adopt for the main analysis.

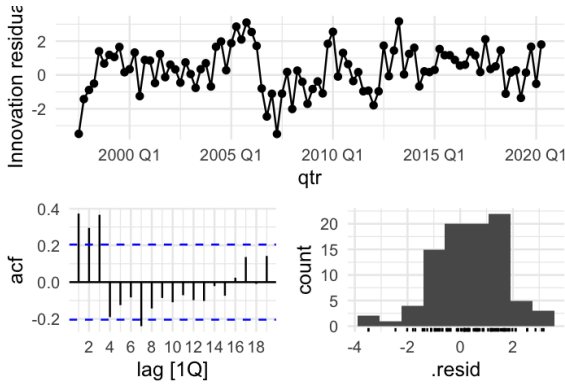
Coefficient	Estimate	(SE)
ϕ_1 (AR(1))	0.9618	0.0461
θ_1 (MA(1))	0.5077	0.1284
θ_2 (MA(2))	0.4283	0.1274
θ_3 (MA(3))	0.4993	0.1355
Φ_1 (SAR(1) at lag 4)	0.6250	0.1926
$\beta(rate_{t-2})$	-0.5884	0.1926
Intercept	77.67	9.155

Table 2: Estimated coefficients of the ARIMAX model with lag $k = 2$.

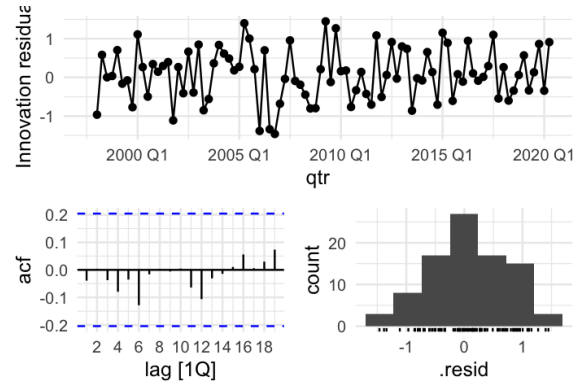
The fitted ARIMAX has the form

$$\Delta y_t = c + \sum_{i=1}^p \phi_i \Delta y_{t-i} + \beta rate_{t-k} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t$$

where β is the coefficient on the lagged rate and the ARMA orders (p, q) are selected by AICc grid search (`stepwise = FALSE`, `Approximation = FALSE`). The estimated coefficients are reported in Table 2, with $\beta = -0.5884$ ($SE = 0.1926$). The negative sign on β is consistent with economic theory, a higher mortgage bond yield raises mortgage costs and dampens subsequent price growth.



(a) Residual plots for the ARIMAX model.



(b) Residual plots for the ARIMAX model with lags.

Figure 4.3a displays the residual diagnostics for the ARIMAX while Figure 4.3b displays the residual diagnostics for the ARIMAX but with lagged rate. For the ARIMAX with lags, the innovation residuals fluctuate around zero with no obvious systematic pattern, the residual ACF lies within the significance bands at all lags, and the histogram is approximately symmetric and kind of bell shaped. The Ljung-Box test returns a $p = 0.915 > 0.05$, confirming that we cannot reject the white-noise hypothesis. For comparison, the regular ARIMAX (with $\beta rate_t$ instead of $\beta rate_{t-k}$) yields strongly autocorrelated residuals, with a Ljung-Box $p < 0.001$, further supporting the use of lagged rate.

4.2 Vector Autoregression (VAR)

While the ARIMAX specification treats the long-term mortgage bond yield as exogenous, this assumption is restrictive in the Danish mortgage bond market, where housing activity directly affect mortgage

issuance and therefore the supply of the bonds whose yield we model. A VAR specification removes this assumption by treating both series as endogenous, and comparing its forecast performance to ARIMAX provides an indirect test of whether the exogeneity assumption is justified in practice.

The model $VAR(p)$ generalizes the univariate AR by letting each variable in the system depend on its own past p lags and on the past p lags of every other variable in the system. In our bivariate setting, the Box-Cox transformed price series and the long-term mortgage bond yield each have their own equation, which allows Granger causality to be tested by examining whether the lags of one variable carry significant weight in the other's equation.

Because both series are $I(1)$, we fit the VAR on first differences. The Schwarz criterion selects $p = 1$, and the Portmanteau test on residuals returns $p = 0.3213$, confirming that the chosen lag order is sufficient to remove residual autocorrelation. The estimated coefficient on the lagged rate in the price equation is highly significant (-0.757 , $p < 0.001$), while the coefficient on the lagged price in the rate equation is not (0.052 , $p = 0.19$), the results is found in Appendix 8.1.1. Causality therefore runs from rates to prices but not back, validating the exogeneity assumption underlying the ARIMAX specification in Section 4.1.2

5 Forecast comparison

To assess out-of-sample performance, all four models are evaluated on the holdout-period 2020 Q3 - 2025 Q4. For the ARIMAX specifications the observed long-term mortgage bond yield is supplied over the test window as a perfect foresight scenario, isolating the predictive contribution of the rate from forecast error in the rate itself. Forecast accuracy is measured by Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) on the Box-Cox transformed scale.

Table 3: Out-of-sample forecast accuracy on the holdout period (2020 Q3 - 2025 Q4), ranked by RMSE. All metrics are computed on the Box-Cox transformed scale.

Model	RMSE	MAE	MAPE
ARIMA	2.38	2.03	2.02
VAR	2.76	2.12	2.10
ARIMAX (lag 2)	10.21	8.81	8.64
ARIMAX (contemp.)	13.74	12.30	12.20

Table 3 ranks the four specifications by RMSE. The univariate ARIMA produces the lowest forecast error across all three metrics, followed closely by the VAR. Both ARIMAX specifications perform substantially worse, with the regular version (`arimax`) ranked last. This pattern suggests that, on this sample, including the long-term mortgage bond yield as an exogeneous regressor does not improve

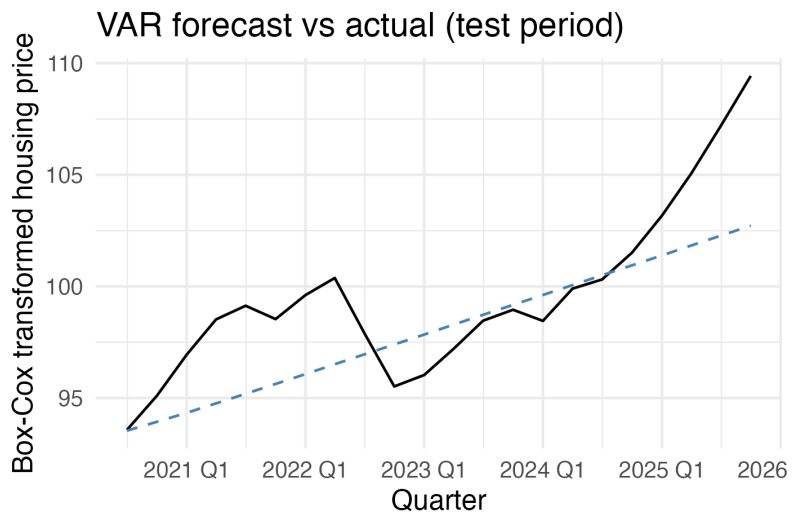


Figure 5.1: VAR forecast (blue dashed) versus realised Box-Cox transformed housing prices (black solid), holdout period 2020 Q3 - 2025 Q4.

6 Structural Break Analysis (QLR)

In a normal setting this section would likely be earlier in the paper, likely after differencing and stationarity testing. A structural break analysis is conducted to assess whether the relationship between housing price

changes and bond yield changes is stable over time. When testing for structural breaks, we are essentially deciding if it is more appropriate to use a full-sample model or a post-break sample. If there is evidence of a break, it

7 Conclusion

8 Appendix

8.1 R Code

8.1.1 Fitted VAR

Table 4: VAR estimation results for equation `price_bc_d`

Parameter	Estimate	Std. Error	t value	Pr(> t)	Sig.
<code>price_bc_d.l1</code>	0.56563	0.08203	6.895	8.17e-10	***
<code>rate_d.l1</code>	-0.75736	0.21664	-3.496	0.000746	***
<code>const</code>	0.14140	0.08363	1.691	0.094434	.

Residual standard error: 0.6988

Multiple R-squared: 0.4149

Adjusted R-squared: 0.4015

F-statistic: 30.85 (p = 7.487e-11)

Table 5: VAR estimation results for equation `rate_d`

Parameter	Estimate	Std. Error	t value	Pr(> t)	Sig.
<code>price_bc_d.l1</code>	0.05166	0.03912	1.320	0.1902	
<code>rate_d.l1</code>	0.24409	0.10332	2.363	0.0204	*
<code>const</code>	-0.07395	0.03988	-1.854	0.0671	.

Residual standard error: 0.3333

Multiple R-squared: 0.07542

Adjusted R-squared: 0.05416

F-statistic: 3.548 (p = 0.03301)

Table 6: Covariance matrix of residuals

	<code>price_bc_d</code>	<code>rate_d</code>
<code>price_bc_d</code>	0.48834	-0.01228
<code>rate_d</code>	-0.01228	0.11107

Table 7: Correlation matrix of residuals

	<code>price_bc_d</code>	<code>rate_d</code>
<code>price_bc_d</code>	1.00000	-0.05275
<code>rate_d</code>	-0.05275	1.00000

8.2 Data

8.2.1 Copenhagen Quarterly House Price Data, 1992Q1–2025Q4